



Metadata is as important as the data itself

Data analytics, by design, is a field that **thrives** on collecting and organizing data. In this reading, you'll learn about metadata and the type of information it can provide. In addition, you'll explore examples of metadata.



Explore a data file by opening any file on your computer or a document in your home or workplace. What is it? Where did it come from? Is it useful? How do you know? This is where metadata comes in to provide a deeper understanding of the data. To put it simply, **metadata is data about data**. In database management, metadata provides information about other data and helps data analysts interpret the contents of the data within a database.

Regardless of whether you're working with a large or small quantity of data, metadata is the mark of a knowledgeable analytics team. Metadata helps people communicate about data across the business and makes it easier to reuse data. In essence, metadata tells the who, what, when, where, which, why, and how of data.

Elements of metadata

Before examining metadata examples, it's important to understand what type of information metadata typically provides:

- File or document type: What type of file or document are you examining?
- Date, time, and creator: When was it created? Who created it? When was it last modified?
- Title and description: What is the name of the item you are examining? What type of content does it contain?
- Geolocation: If you're examining a photo, where was it taken?
- Tags and categories: What is the general overview of the item that you have? Is it indexed or described in a specific way?
- Who last modified it and when: Were any changes made to the file? If yes, when were the most recent modifications made?
- Who can access or update it: If you're examining a dataset, is it public? Are special permissions needed to customize or modify it?

Examples of metadata

In today's digital world, metadata is everywhere! Here are some examples—with accompanying images—of where you might find metadata.

Photos

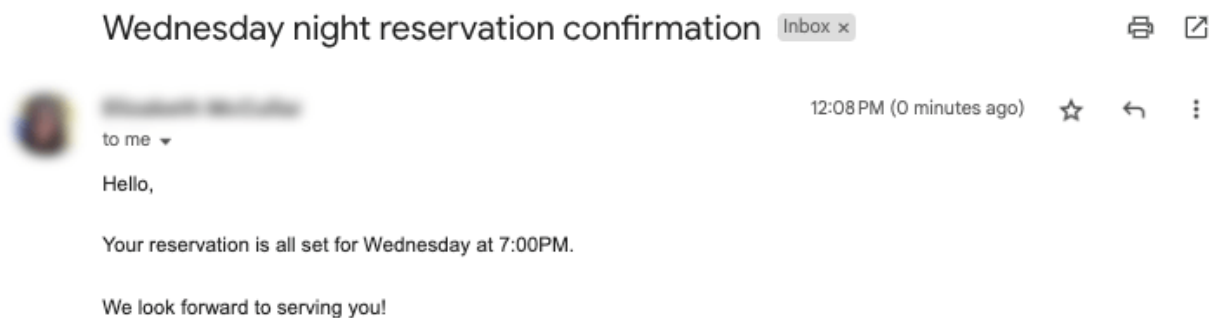
Whenever a photo is captured with a camera, metadata such as filename, date, time, geolocation, and the type of device on which it was taken are gathered and saved with it. The metadata of the following photo is displayed as a pop-up alongside the photo.



Image with accompanying Information pop-up that displays its description, the date and time the image was taken, its size, the device on which it was taken, and an option to add the geolocation of the image.

Emails

When an email is sent or received, it contains metadata such as subject line, sender, recipient, date sent, and time sent.



Emails also contain hidden metadata that includes server names, IP addresses, HTML format, and software details. This image includes hidden email metadata such as the message ID and when the email was created.

Original Message

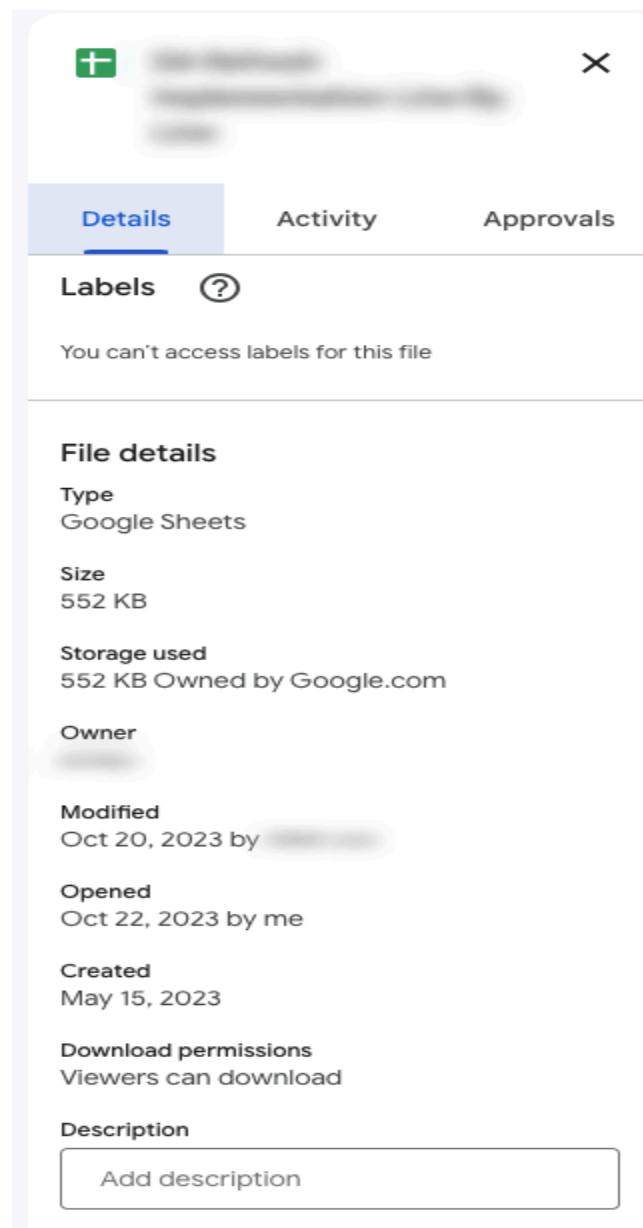
Message ID	<CACdaXyi4ZA4OwhMc6VTBkm9O_fE+GU_MQ9ZPu4cKdG+gbUW6xg@mail.gmail.com>
Created at:	Mon, Oct 23, 2023 at 12:08 PM (Delivered after 12 seconds)
From:	[Redacted]
To:	[Redacted]
Subject:	Wednesday night reservation confirmation
SPF:	PASS with IP 209.85.220.65 Learn more
DKIM:	'PASS' with domain sweetrush.com Learn more
DMARC:	'PASS' Learn more

Hidden metadata from an email that includes Message ID, creation date, the recipient, the sender, the subject line, the SPF, DKIM, and DMARC.

Spreadsheets and electronically created documents

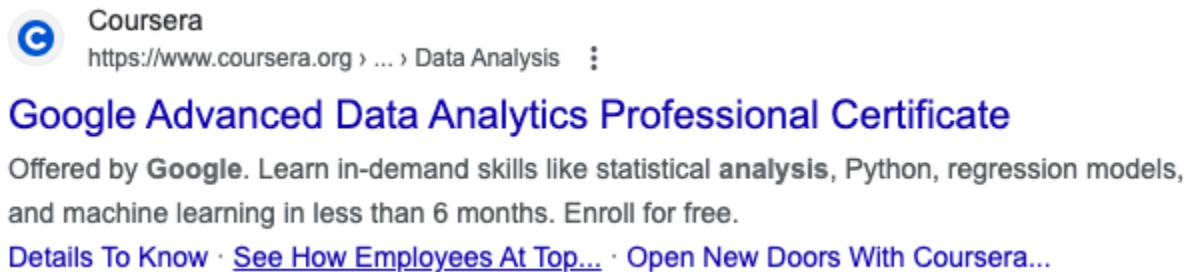
Spreadsheets and documents are already filled with a considerable amount of data, so it's no surprise that they also include metadata such as title, author, creation date, number of pages, and user comments. Additionally, spreadsheet metadata includes tab names, tables, and columns. In the following example, the image demonstrates the metadata for an electronically created Google Sheet:

The metadata of a Google Sheet including the title, type, size, storage used, owner, the last person who modified the document, the last person who opened the document, when it was created, download permissions, and an option to include a description.



Websites

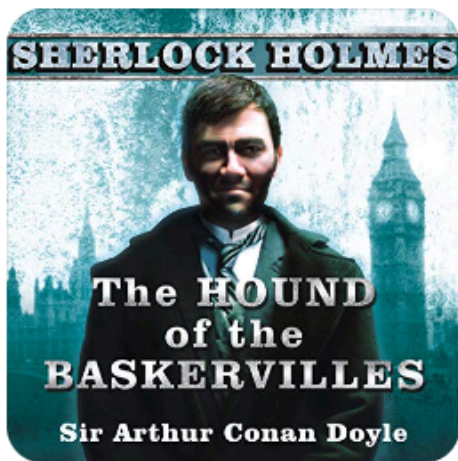
Every web page has a number of standard metadata fields such as tags and categories, the site creator's name, web page title and description, and time of creation. Results of search engine queries that you might make on a daily basis are metadata!



A search engine result that includes meta title as Google Advanced Data Analytics Professional Certificate. Under the website hyperlink is the Meta description that says: Offered by Google. Learn in-demand skills like statistical analysis, Python, regression models, and machine learning in less than 6 months. Enroll for free.

Books and audiobooks

Non-digital items can have metadata, too! Every book has standard metadata that will inform you of its title, author's name, a table of contents, publisher information, copyright description, index, and a brief description of the book's contents. Audiobook metadata also includes this data, as well as metadata specific to the audiobook such as narrator and recording length.



▶ Sample

The Hound of the Baskervilles

A Sherlock Holmes Novel

By: [Arthur Conan Doyle](#)
Narrated by: [Simon Prebble](#)
Series: [Sherlock Holmes](#), Book 5
Length: 6 hrs and 41 mins
Unabridged Audiobook
Release date: 11-30-09
Language: English
Publisher: [Tantor Audio](#)
★ ★ ★ ★ ☆ 4.4 (843 ratings)

1 CREDIT

Audiobook's metadata including the title of the audiobook, author, narrator, its length, the release date, language the audiobook is read in, its published, and its rating score.

Key takeaways

Metadata can be found in photos, emails, spreadsheets, websites, and much more! In your daily life, **you use metadata to stay organized**. As a data analyst, you'll use metadata to understand the content and context of your data, as well as how it's structured. Metadata provides data analysts with information about a data's file type, title, geolocation, who created it, who last modified it, and who has access to it. As a data analyst, it's important to keep accurate records of metadata to ensure that you are able to find, use, preserve, and reuse data in the future. Remember, it will be your responsibility to manage and make use of data in its **entirety**; metadata is as important as the data itself.